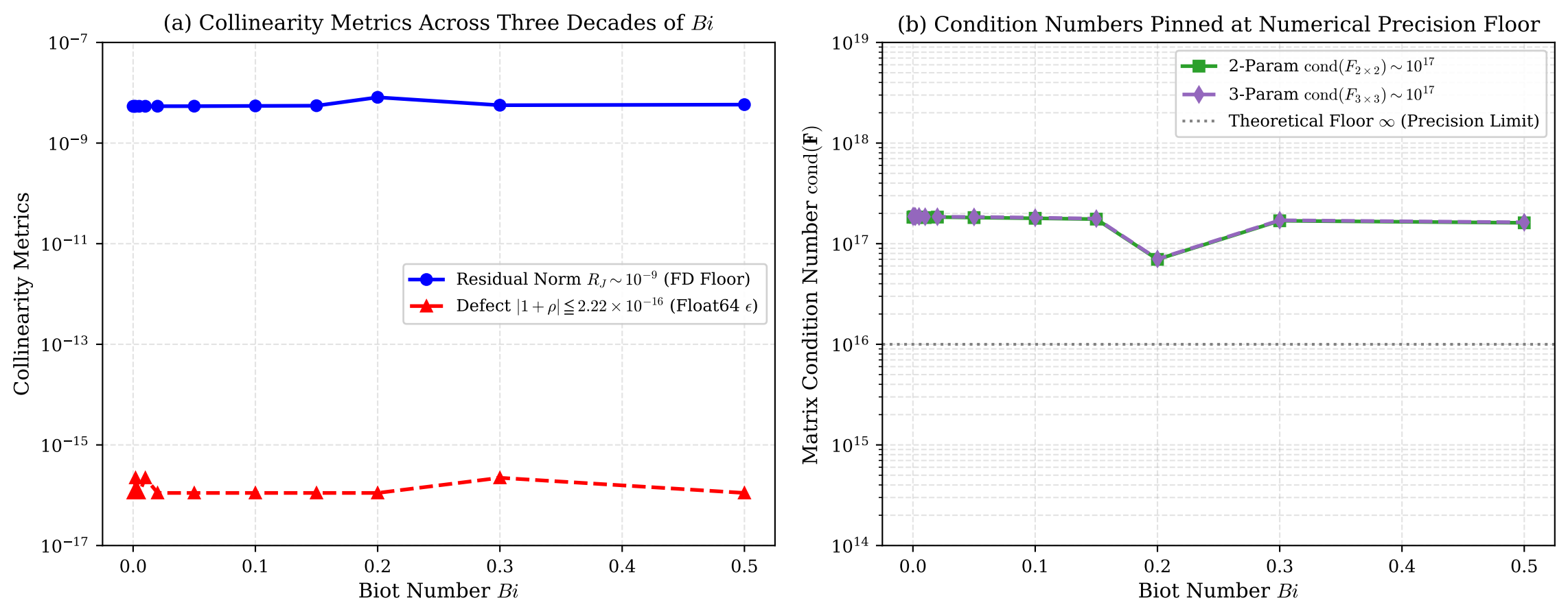
(a) Collinearity Metrics Across Three Decades of $Bi$
Collinearity Metrics
$10^{-7}$
$10^{-9}$
$10^{-11}$
$10^{-13}$
$10^{-15}$
$10^{-17}$
Residual Norm $R_J \sim 10^{-9}$ (FD Floor)
Defect $|1+\rho| \leqq 2.22 \times 10^{-16}$ (Float64 $\epsilon$)
0.0
0.1
0.2
0.3
0.4
0.5
Biot Number $Bi$
(b) Condition Numbers Pinned at Numerical Precision Floor
Matrix Condition Number $\mathrm{cond}(\mathbf{F})$
$10^{19}$
$10^{18}$
$10^{17}$
$10^{16}$
$10^{15}$
$10^{14}$
2-Param $\mathrm{cond}(F_{2\times 2}) \sim 10^{17}$
3-Param $\mathrm{cond}(F_{3\times 3}) \sim 10^{17}$
Theoretical Floor $\infty$ (Precision Limit)
0.0
0.1
0.2
0.3
0.4
0.5
Biot Number $Bi$